NILAM — End-to-end System Architecture

Kerala land intelligence → AI-assisted house design → district-aware construction budget

LOCAL-FIRST PRODUCT · CLOUD DATA ON DEMAND

01 · INPUTS

User & live evidence

Where to inspect and what the user needs

Map & requirement input
- Search, click, polygon or device location
- Centre point + radius for land discovery
- Optional text or English voice preferences

Live geospatial evidence

Google Earth Engine
Terrain · land cover · vegetation · water

OpenStreetMap + Kerala hazard layers
Buildings · roads · amenities · flood · landslide

Google Maps, Places and Street View

House & budget preferences

Plot dimensions · floors · room programme
Style · relationships · parking · setbacks
District · finish level · design refinements

USED AFTER SITE SELECTION

site / radius

02 · LAND SUITABILITY

Geospatial screening & multimodal AI

Site evidence, hard screening, prediction and explanation

Site feature service & candidate search

Build 38 mapped terrain, climate, soil, water, hazard, access, utility and amenity features.
For radius search, sample fresh grid cells and reject steep, rough, flooded, built-up or building-adjacent cells.

TABULAR

TabPFN 3.5
38 site features
4 suitability classes
Primary expert · 92% blend

VISION

TerraMind Base
Dry + monsoon Sentinel-2
224 × 224 × 12 per season
Fine-tuned expert · 8% blend

INTENT

Laya router
Preference extraction
+ deterministic numbers
Distil-Whisper for speech

Decision engine

Hard safety/open-land gates → calibrated class-probability fusion → ordinal suitability percentage
Conditional permutation SHAP explains the final result; amenity and evidence services add local context.

Land intelligence output

Final score + suitability band · feature drivers · ranked flat/open candidates · amenities · Street View

selected land

03 · HANDOFF

Site context

One shared contract

SITE_CONTEXT

Selected site

Latitude
Longitude
District
Suitability %

Passed into the home brief and cost estimator.

site context

house requirements

04 · ARCHAI2

House generation & refinement

Natural-language brief to validated 2D/3D geometry

Qwen 3.5 9B · local LM Studio

Converts the user's brief and refinements into schema-constrained house requirements.
Language interpretation only; geometry must pass deterministic validation.

Spatial intent compiler

Room programme · adjacency
Plot shape · floors · setbacks
Canonical structured specification

Candidate generation

HouseGAN++ proposal when eligible
+ procedural constrained optimizer
Multiple layout candidates

Independent validation & selection

Rejects collisions and infeasible geometry; checks access, circulation, room area, adjacency and setbac
Only validated structured geometry can become the saved revision.

Canonical geometry

Rooms · walls · doors · windows
Stairs · footprint · site boundary
One source for 2D, 3D and exports

Projects & revisions

FastAPI services + SQLite history
Chat refinements + candidate choice
Next.js workspace

saved revision

05 · COST

Kerala cost calculator

Concept budget from the saved design

Estimator inputs

Saved house geometry
Kerala district from Nilam
Chosen finish level

Cost engine

Union room polygons by floor
+ wall / semi-open allowances

Kerala finish-rate bands
× district material/labour factor

+ complexity and room allowances
+ 12% planning reserve

Concept-stage budget

Low / planning / high range
Area and cost breakdown
District factor + assumptions

End-user outputs

One continuous Nilam journey, with each ou

Assess a site

Suitability percentage · evidence drivers · SHAP explanation
Nearby amenities · Street View · saved location

Find suitable land

Ranked flat/open candidate points within the chosen radius
Preference match · site detail · Street View inspection

Design and refine the home

Validated 2D floor plan · interactive 3D model · revision history
SVG / PNG / PDF / GLB / GLTF exports

Estimate construction cost

Kerala district-adjusted planning budget and range
Transparent quantities, assumptions, exclusions and source links

model registry / local artifacts

Offline evidence build, training & evaluation

Executed ahead of runtime; produces versioned data and model artifacts loaded by the land service

Authoritative and public data

SRTM · CHIRPS · ERA5-Land · Dynamic World
MODIS NDVI · JRC water · OpenLandMap
GSI / KSDMA / UNEP hazards · Kerala OSM PBF

Statewide academic cohort

1,400 points · 100 per Kerala district
38 tabular features + weak-label class / confidence
1,400 paired dry/monsoon Sentinel-2 chip files

Model fitting

Fit TabPFN on mapped features
Fine-tune TerraMind vision head + final two encoder blocks
Whole-district train / validation / test split

Calibration & evaluation

Validation-selected 92% / 8% geometric pool
Accuracy · macro F1 · precision · recall · log loss
Fusion and classical-model ablations

Versioned runtime artifacts

TabPFN estimator · TerraMind checkpoint / head
Fusion configuration · feature schema · metrics
Loaded locally by Nilam FastAPI

Runtime: React + Google Maps UI · Nilam FastAPI (:8000) · ArchAI2 FastAPI (:8001) · Next.js workspace (:3000) · local SQLite + caches

System boundary: NILAM